

Sobolev Spaces and the Direct Method in the Calculus of Variations

Sarthak Dassarma (template from Professor Hartglass)



Santa Clara University

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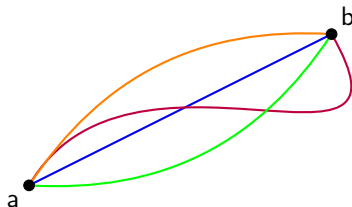
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- 5 Finding the Minimizer

The Shortest Curve Between Two Points

Given two points

$$a, b \in \mathbb{R}^n, \gamma : [0, 1] \rightarrow \mathbb{R}^n, \gamma(0) = a, \gamma(1) = b$$

what is the shortest curve joining them?

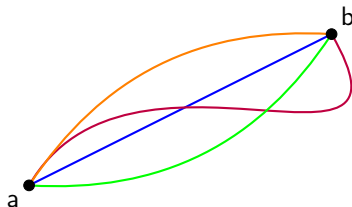


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The length of the curve is

$$L(\gamma) = \int_0^1 |\gamma'(t)| dt$$

Question:

$$\min_{\gamma} L(\gamma)?$$

A Problem in the Calculus of Variations

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Suppose

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Does a limit curve exist? If it exists, is it still differentiable?

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Weak Derivatives and the Need for a Larger Function Space

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Recall the integration by parts formula:

$$\int_{\Omega} u'(x) \varphi(x) dx = - \int_{\Omega} u(x) \varphi'(x) dx$$

for all test functions

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The right-hand side still makes sense even when u is not classically differentiable.

Definition of Weak Derivative

Definition

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The derivative is now defined distributionally rather than pointwise.

Example: A Weak Derivative

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Sharp corners are okay, but we *cannot have discontinuous jumps*.

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For the remainder of this talk we focus on the case $p = 2$.

Hence

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The Sobolev Norm

Definition

For $u \in W^{1,2}(\Omega)$, define

$$\|u\|_{W^{1,2}} = \left(\|u\|_{L^2}^2 + \|Du\|_{L^2}^2 \right)^{1/2}$$

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Fact: $W^{1,p}(\Omega)$ equipped with the Sobolev norm is a Banach space. Even better, $W^{1,2}(\Omega)$ is a Hilbert space, so of course it is complete!

Weak Convergence

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Notation:

$$u_k \rightharpoonup u$$

Compactness

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Theorem

*Every bounded sequence in $W^{1,2}(\Omega)$ has a weakly convergent subsequence.
 $|u_k|_{W^{1,2}} \leq C$ implies that some subsequence satisfies $u_{k_j} \rightharpoonup u$.*

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Closed, bounded sets are generally not compact in the norm topology.
However, closed convex sets agree in both the norm and weak topologies. (This is essential for minimization!)

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Why energy instead of length? Because E behaves much better with respect to weak convergence in $W^{1,2}$.

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Weak compactness gives Step 2.

Weak lower semicontinuity gives Step 3.

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If $\gamma_k \rightharpoonup \gamma$ in $W^{1,2}$, then

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Therefore energy cannot suddenly decrease in the weak limit.

Existence of a Minimizer

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Hence, $E(\gamma) = \inf_{\mathcal{A}} E$, and a minimizer exists.

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A Lower Bound for the Energy

Let $\gamma : [0, 1] \rightarrow \mathbb{R}^n$ be a curve connecting $\gamma(0) = a$ to $\gamma(1) = b$. We define the energy as:

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Applying Cauchy–Schwarz:

$$|b - a|^2 = \left| \int_0^1 \gamma'(t) dt \right|^2 \leq \left(\int_0^1 |\gamma'(t)|^2 dt \right) \left(\int_0^1 1^2 dt \right)$$

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Since $\int_0^1 1 dt = 1$, we obtain the sharp lower bound:

$$|b - a|^2 \leq E(\gamma)$$

The Straight Line Attains the Bound

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Therefore, the energy of this straight line is:

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Thus, the straight line is a minimizer of the energy.

From Energy to Length

By Cauchy–Schwarz, we can relate length to energy:

$$(L(\gamma))^2 = \left(\int_0^1 |\gamma'(t)| \cdot 1 \, dt \right)^2 \leq \left(\int_0^1 |\gamma'(t)|^2 \, dt \right) \left(\int_0^1 1^2 \, dt \right) = E(\gamma)$$

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Therefore, the straight line is also the shortest curve joining a and b .

Conclusion

Sobolev spaces provide:

- weak derivatives
- completeness
- weak compactness
- existence of minimizers via the Direct Method